

CART

- Classification and Regression Tree
 - Key idea:
 - Only Two Children
 - Some goodness criterion to split (Typically minimize Ginni Index based)
 - $\sum p_t(1 - p_t)$
 - Pruning to avoid overfitting (Typically Information Gain)

In general,

Decision trees, good for explaining decisions

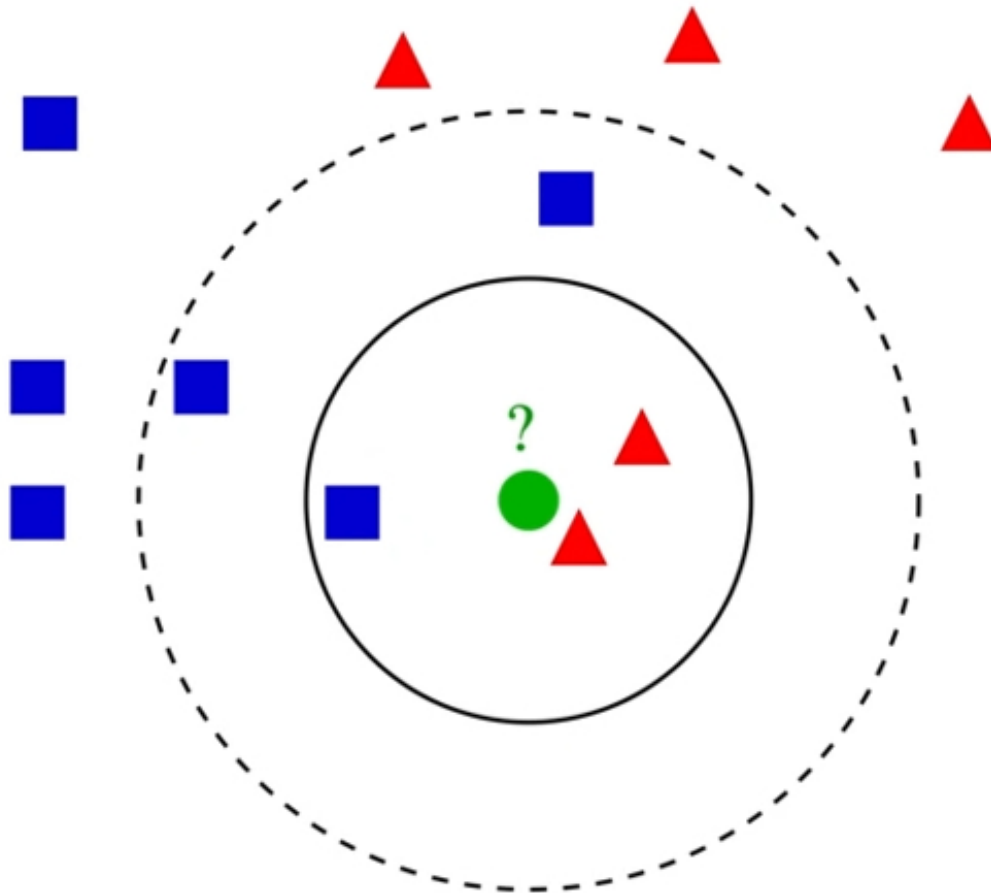
However, very sensitive to noise in the data, small change in the data can lead to a very different different tree

Bayes vs Naïve Bayes vs Decision Tree

- Optimal
- This is ultimate goal
- practically difficult to learn Bayes classifier directly from the data
- Curse of Dimensionality
- Easy to build
- Not all features independent
- Explainability is poor
- Easy to build
- Explainability is good
- Overfitting
- sensitivity

kNN

- 'k' nearest neighbours
- Data set $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$
- Given a point x , find out k nearest point from D to x
- E.g., say $|x_{\sigma(1)} - x| \leq |x_{\sigma(2)} - x| \leq \dots \leq |x_{\sigma(n)} - x|$
- Collect labels of $x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(k)}$
- $\hat{y}$ = the most frequently occurring label in $x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(k)}$



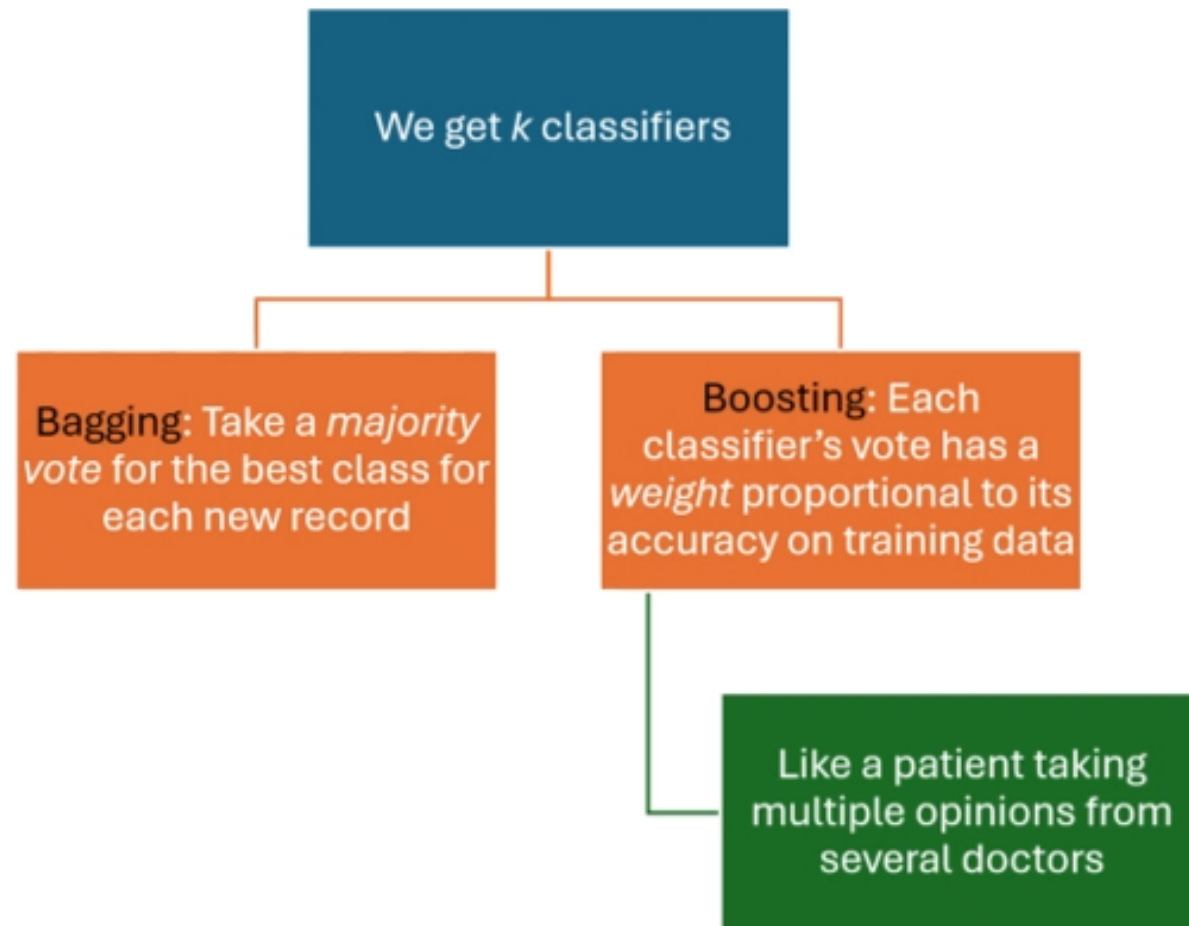
What will be label if $k = 3$?

What will be label if $k = 5$?

- How to choose right k ?
- On the given data,
 - estimate error on different k values
 - select the one with least error
 - Choose the k value where a small decrease in k causes a large increase in error and increase in k results small decrease in the error

- Running time, linear in $n \times d$
- As $n \rightarrow \infty$, 1-NN error rate is at most $2 \times \textit{Bayes error rate}$
- How to measure **nearness**?
 - If features are continuous, Euclidian metric
 - Discrete features: hamming distance
- Drawback
 - Frequent classes dominate

Combining Classifiers





Clustering

Birds of a feather flock
together

Applications

- Targetting similar people or objects
 - Student tutorial groups
 - Hobby groups
 - Health support groups
 - Customer groups for marketing
 - Organizing e-mail
- Spatial clustering
 - Exam centres
 - Locations for a business chain
 - Planning a political strategy
- Two Types of algorithms
 - Hard Clustering: Each point is assigned to some cluster
 - Soft Clustering: Each point is assigned probabilities of being assigned to each cluster

The Clustering Problem

- *Find groups of similar records.*
- Need a function to compute similarity, given 2 input records
- Unsupervised learning

Outlook	Temp (°F)	Humidity (%)	Windy?
sunny	75	70	true
sunny	80	90	true
sunny	85	85	false
sunny	72	95	false
sunny	69	70	false
overcast	72	90	true
overcast	73	88	true
overcast	64	65	true
overcast	81	75	false
rain	71	80	true
rain	65	70	true
rain	75	80	false
rain	68	80	false
.	70	80	6.1

Hierarchical Clustering

- Agglomerative (e.g. AGNES):

- Start: Each point in separate cluster
- Merge 2 closest clusters
- Repeat until all records are in a single cluster
- Computationally Expensive
- Better at handling outliers

- Divisive (e.g. DIANA)

- Start: All points in 1 cluster
- Find most extreme points in each cluster.
- Regroup points based on closest extreme point
- Repeat until each record is in its own cluster
- Outliers may disrupt the

These are called **Connectivity-based Clustering**



a

b

c

d

e

f

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Measurement of Similarity

❑ Nominal (categorical) variables

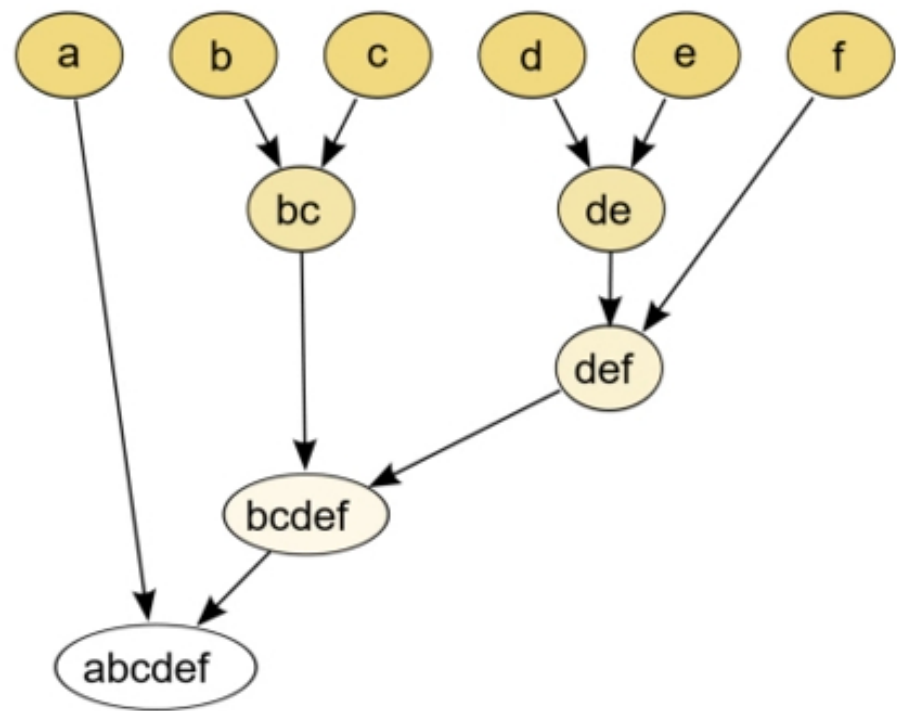
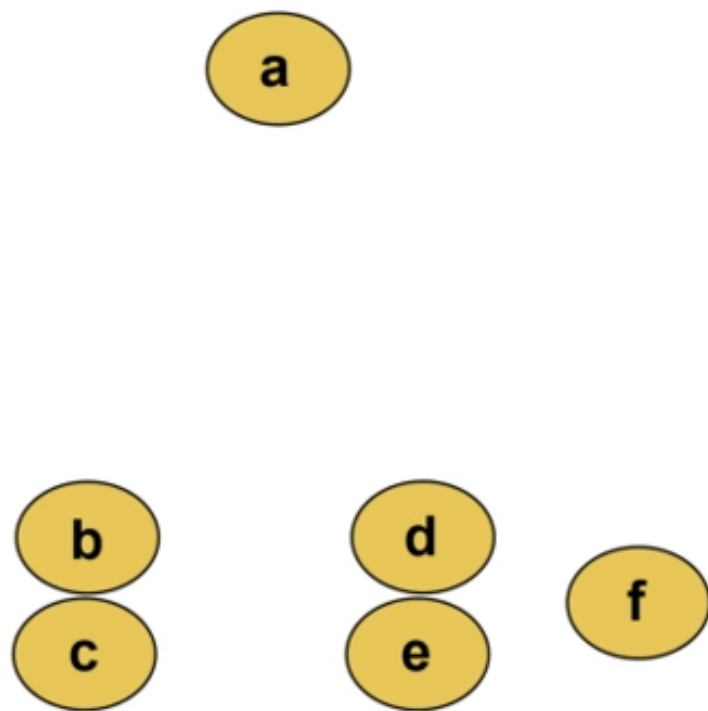
- $d(x,y) = 1 - m/n$
- m = no of matches among n attributes, or
 m = sum of weights of matching attributes, and n is the sum of weights of all attributes

❑ Numeric variables

- Euclidean, Manhattan, Minkowski,...
- Ordinal

❑ Above are examples

- Similarity is ultimately application dependent
- Requires various kinds of preprocessing
- Scaling: Convert all attributes to have same range
 - z-score: $z = (\text{value} - \text{mean})/m$
 - where m is the mean absolute deviation



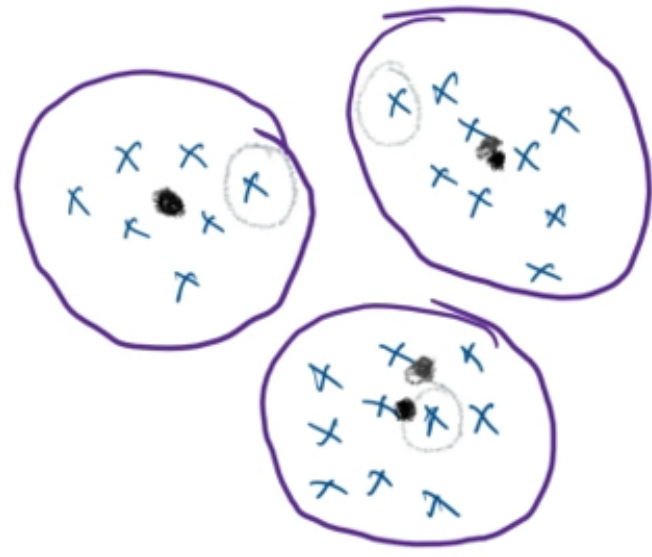
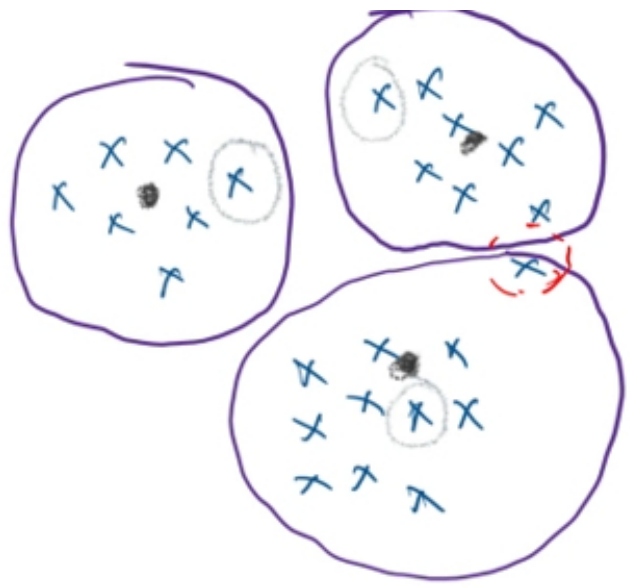
What is goal?

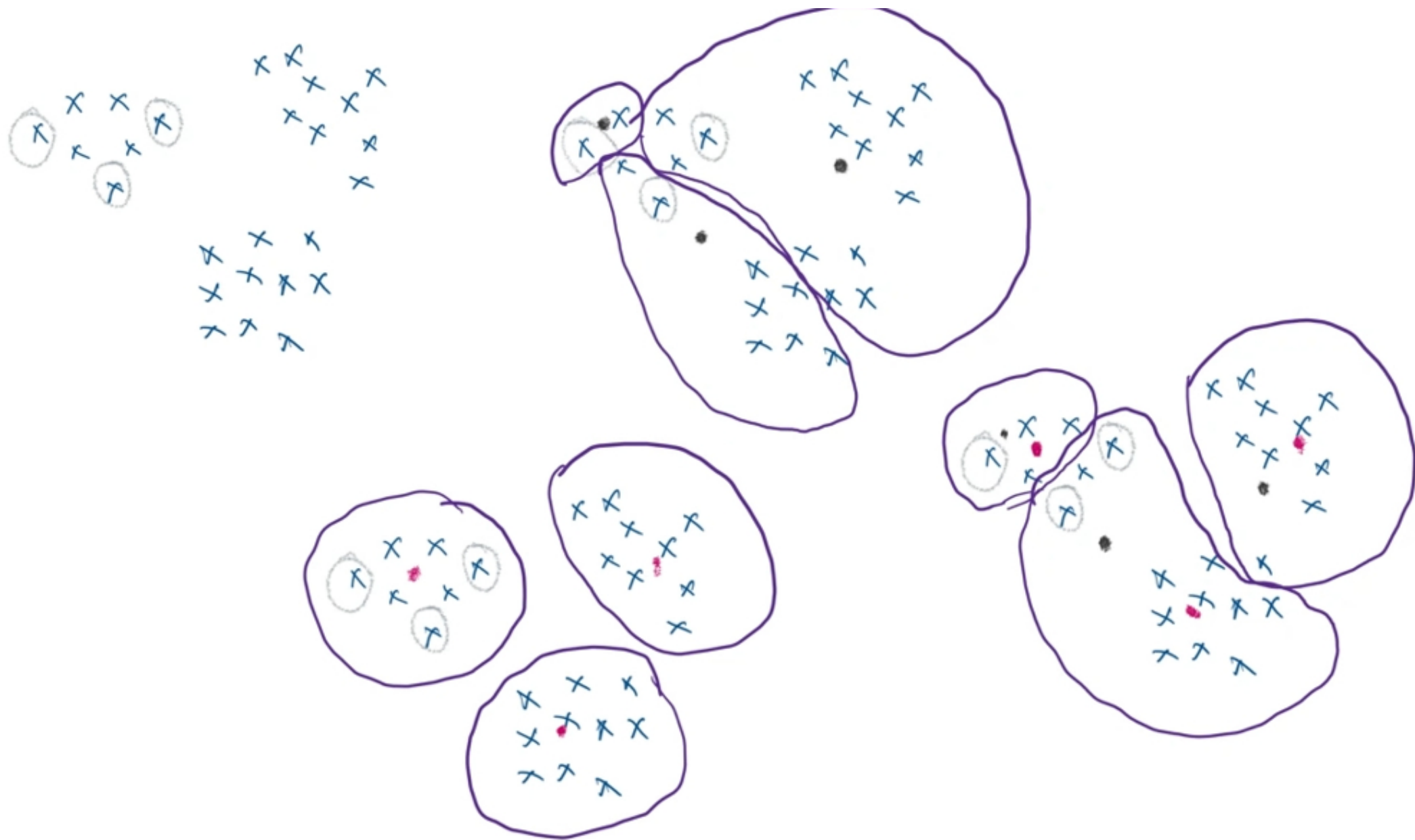
- Similar points go together
- That is, partition the points into k sets
 - Within a cluster, points are similar
 - Variance is minimized

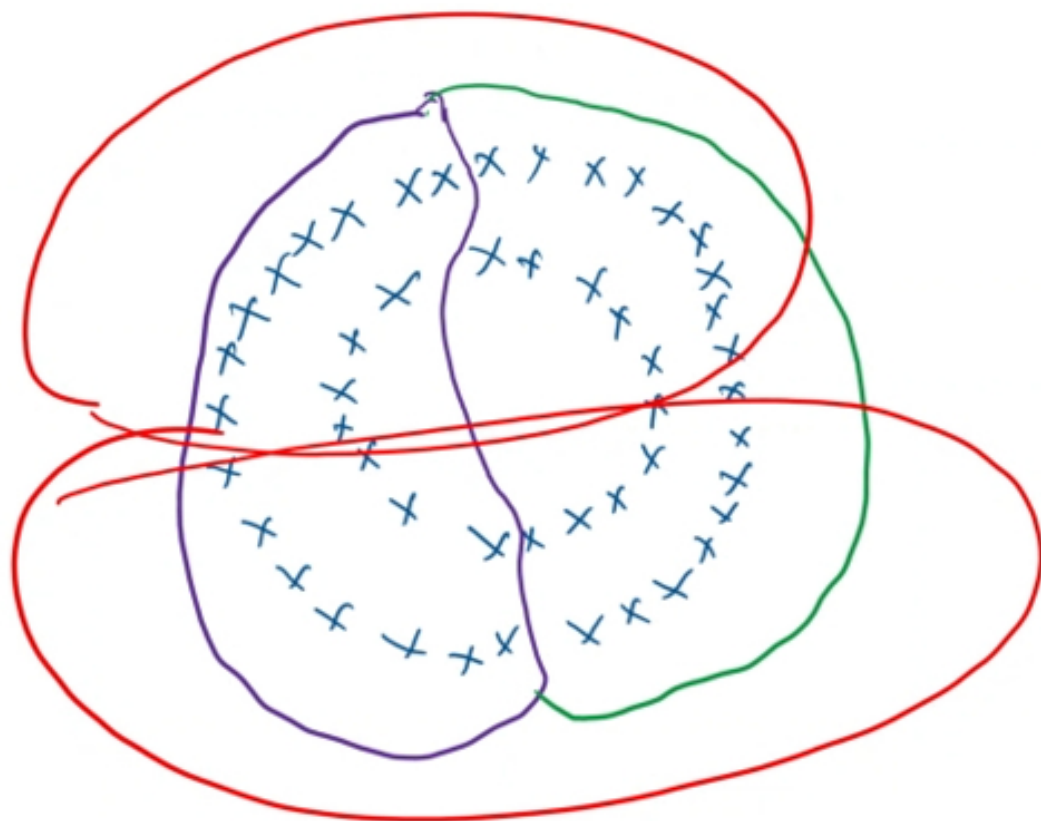
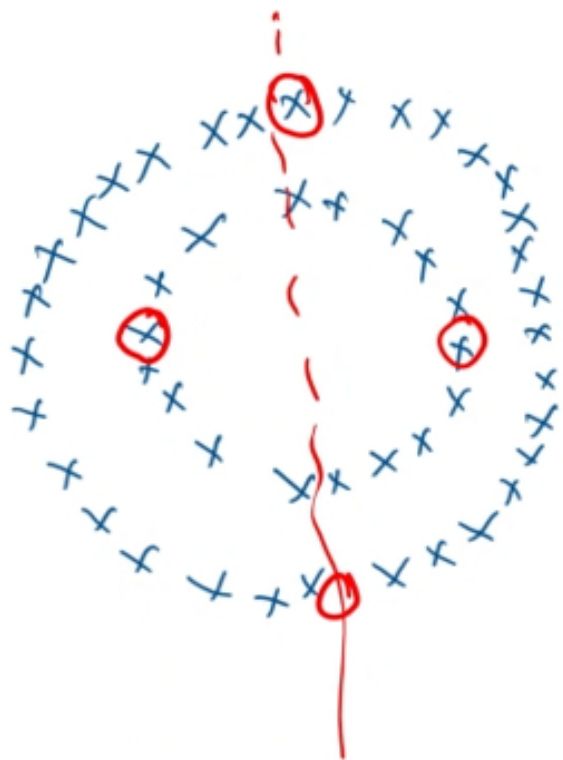
k -means

- Randomly select k points as centers
- Assign each point to the cluster closest to these centers
- Compute mean of all the points in the cluster
- Call these points as new centers of the cluster
- Continue still no further improvement

This technique falls under **Centroid-based Clustering**







Which k ? Which Algorithm?

- Project on 2D/3D and determine if they are visually nicely separated
- **Silhouette Score**
 - $S(i)$ = Based on average distance with points in the same cluster $a(i)$ vs smallest distance to the points not in the cluster ($b(i)$)
 - $S(i) = \frac{b(i)-a(i)}{\max\{a(i),b(i)\}}$
 - Average $S(i) \in [-1, 1]$, higher the better)
- **Davies-Bouldin Index**
 - Captures compactness of a cluster with its separation across clusters
 - $DB = \frac{1}{k} \sum_i \max_j \frac{\Delta X_i + \Delta X_j}{\delta(X_i, X_j)}$ Δ : Intracluster distance δ : intercluster distance